Grade 11 – Term 1
C1 Practice Activity 1 with Solutions

Contents

Evaluated criterion

- C1: Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.

1 Introduction

A **random variable** assigns a numerical value to the outcome of a random process. A discrete random variable takes separate, countable values (often counts such as $0, 1, 2, \ldots$). A continuous random variable can take any real value in an interval, such as a waiting time or price. Its **support** is the set of possible values.

For a discrete variable, the probability mass function (PMF) $p_X(x) = P(X = x)$ gives probability at each possible value, and the masses sum to 1. For a continuous variable, the probability density function (PDF) $f_X(x)$ is a height, not a point probability: $P(X = x) = 0$, while interval probability is area, $P(r \leq X \leq s) = \int_r^s f_X(x)\,dx$. A valid density is nonnegative and has total area 1. In both cases the cumulative distribution function (CDF)

$$F_X(x) = P(X \le x)$$

records the probability accumulated up to x; it is a sum of PMF values or an integral of the PDF.

1.1 Bernoulli Distribution

A Bernoulli variable represents *one* trial with two outcomes, coded success (1) and failure (0). Clues are "one selected account," "yes/no," or "paid/defaulted." Its support is $\{0,1\}$ and its parameter is the success probability $0 \le p \le 1$:

$$P(X = x) = p^x(1-p)^{1-x}, \quad x \in \{0,1\}, \qquad F_X(x) = \begin{cases} 0, & x < 0, \\ 1-p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

The PMF assigns the two outcome probabilities; the CDF gives the probability of a coded outcome no greater than x. For example, $X = 1$ may mean that one randomly selected invoice is overdue.

1.2 Binomial Distribution

A binomial variable counts successes in a *fixed* number n of independent Bernoulli trials having the same success probability p. Its recognition clues are a stated sample size, binary outcomes, independence, and constant p. Its support is $\{0, 1, \ldots, n\}$ and its parameters are n and p:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, \ldots, n, \qquad F_X(x) = \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k (1-p)^{n-k},$$

where the CDF is 0 below 0 and 1 at or above n. The PMF gives the probability of exactly x successes; the CDF gives that of at most x. An example is the number of renewed subscriptions among n independently contacted customers.

1.3 Geometric Distribution

Here X counts the number of independent Bernoulli trials *required to obtain the first success*; thus this activity uses support $\{1, 2, 3, \ldots\}$. Look for repeated independent attempts, constant success probability p, and stopping at the first success:

$$P(X = x) = (1-p)^{x-1} p, \quad x = 1, 2, \ldots, \qquad F_X(x) = \begin{cases} 0, & x < 1, \\ 1 - (1-p)^{\lfloor x \rfloor}, & x \ge 1. \end{cases}$$

The PMF describes $x - 1$ failures followed by a success; the CDF describes success occurring by trial x. For example, X can be the number of sales pitches until the first order.

1.4 Poisson Distribution

A Poisson variable counts events in a fixed time, distance, area, or other interval when events occur separately at an approximately constant average rate $\lambda > 0$. Clues are an event count, a fixed interval, and a stable rate. Its support is $\{0, 1, 2, \ldots\}$:

$$P(X = x) = e^{-\lambda}\frac{\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots, \qquad F_X(x) = e^{-\lambda}\sum_{k=0}^{\lfloor x \rfloor}\frac{\lambda^k}{k!}.$$

The PMF gives the chance of exactly x events and the CDF the chance of no more than x. A call centre might use it for the number of incoming calls in a fixed five-minute period.

1.5 Uniform Distribution

A uniform variable treats all values in a bounded interval $[a, b]$ as having equal density. A stated minimum and maximum plus "equally plausible anywhere" are decisive clues. Its support is $[a, b]$, with parameters $a < b$:

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise}, \end{cases} \qquad F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$

The constant PDF makes equal-length subintervals equally probable; the CDF is the fraction of the support lying at or below x. A courier arrival time known only to be equally plausible from 2:00 to 2:30 p.m. is an example.

1.6 Triangular Distribution

A triangular variable has minimum a, maximum b, and mode (most likely value) c, where $a < c < b$. Density rises linearly toward c and falls linearly afterward—the key distinction from one straight linear density across the whole support. Its parameters are a, c, b and support is $[a, b]$:

$$f_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{2(x-a)}{(b-a)(c-a)}, & a \le x \le c, \\ \dfrac{2(b-x)}{(b-a)(b-c)}, & c < x \le b, \\ 0, & x > b, \end{cases}$$

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{(x-a)^2}{(b-a)(c-a)}, & a \le x \le c, \\ 1 - \dfrac{(b-x)^2}{(b-a)(b-c)}, & c < x \le b, \\ 1, & x > b. \end{cases}$$

The PDF expresses increasing then decreasing plausibility; its accumulated area is the CDF. A project cost with an expert minimum, maximum, and most likely estimate is a common business example.

1.7 Linear Distribution

A linear distribution has a PDF $f_X(x) = mx + d$ throughout a bounded support $[a, b]$, and 0 outside it. The constants must satisfy $mx + d \geq 0$ on $[a, b]$ and

$$\int_a^b (mx + d)\, dx = \frac{m}{2}(b^2 - a^2) + d(b - a) = 1.$$

Its support is $[a, b]$ and its parameters may be written a, b, m, d subject to these restrictions. The CDF is

$$F_X(x) = \begin{cases} 0, & x < a, \\ \frac{m}{2}(x^2 - a^2) + d(x - a), & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

Clues specify one density line increasing or decreasing across the entire interval, without an interior mode. The PDF gives relative concentration and the CDF its accumulated area. For instance, longer checkout times from 0 to 8 minutes may have density $f(t) = t/32$, increasing linearly over the whole interval.

1.8 Normal Distribution

A normal variable models approximately symmetric, bell-shaped observations clustered around a centre, with values farther away increasingly uncommon. Its support is all real numbers and its parameters are mean μ and standard deviation $\sigma > 0$:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right], \qquad F_X(x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Here Φ is the standard normal CDF (it is normally read from technology or a table). The PDF describes the bell's relative concentration and the CDF gives area to the left. A large firm's many employee commute times may be approximately normal when they are symmetric around a mean.

1.9 Summary

Type	Distribution	Typical situation
Discrete	Bernoulli	One situation with two possible outcomes.
Discrete	Binomial	Counting how many times an outcome occurs across a fixed number of comparable cases.
Discrete	Geometric	Counting how many attempts are required until the first desired outcome.
Discrete	Poisson	Counting how many separate events occur during a fixed period or exposure.
Continuous	Uniform	A bounded quantity for which no part of the possible range is expected more often than another.
Continuous	Triangular	A bounded quantity with a practical minimum, maximum, and one most common value.
Continuous	Linear	A bounded quantity whose observations become steadily more or less common across its range.
Continuous	Normal	Values concentrated around a typical value, with increasingly uncommon observations farther from it on either side.

2 Discrete Probabilities

Problems:

2.1 Bogotá Venture Decisions

Consider the following four situations.

1. A Bogotá fintech is monitoring one transfer to a freelance designer who needs the funds for this month's studio rent. The transfer will either clear during the banking window or be rejected, and no other result is recorded. The variable X_1 records which of those two outcomes occurs so the platform can decide whether to release the funds or begin a review.
2. A meal-preparation venture in Chapinero is calling offices to secure its first recurring lunch contract. Staff use the same current offer with one office after another and stop calling as soon as an office books a tasting; recent calling rounds show comparable responses from offices approached this way. The variable X_2 records how many calls are needed through that first booking, which determines whether this sales channel is affordable.
3. A flower exporter near Bogotá must decide whether to release a shipment without costly re-sorting. Inspectors select exactly 40 stems packed on the same line and record for each one whether it meets export grade; recent batches show a consistent pass experience across stems. The variable X_3 counts the stems that pass, giving the manager evidence for the release decision.

4. A Bogotá concert promoter must staff its customer-assistance team after announcing a schedule change. Refund requests are logged separately during one hour, and records from comparable announcement periods show activity spread fairly steadily through the hour rather than arriving on a timetable. The variable X_4 counts the requests logged in that hour so the promoter can balance response times against labour cost.

Questions/Tasks.

C1 Characterise each random variable and assign it to the appropriate probability distribution.

C1 - Characterise the random variables.

All four variables are discrete.

1. X_1 is **Bernoulli**: one transfer is one trial with exactly two outcomes, clearing or rejection.
2. X_2 is **geometric**: independent, equal-probability calls continue until the first appointment.
3. X_3 is **binomial**: it counts passes in a fixed set of 40 independent binary inspections with constant pass probability.
4. X_4 is **Poisson**: it counts separate events in a fixed hour under an approximately stable rate.

2.2 Creative and Digital Markets

Consider the following four situations.

1. A Bogotá podcast studio runs a two-day sponsorship campaign for a series about Colombian creative businesses. Enquiries appear as separate messages throughout comparable campaigns, with no recurring surge during the window and no scheduled sender list. The variable X_1 counts the enquiries received in those two days so the studio can staff replies and judge whether the campaign is commercially worthwhile.
2. A social-commerce seller in Suba is testing 25 paid advertisements before committing more working capital. Each advertisement uses the same offer with a comparable audience, and the seller records whether it leads to a purchase during the campaign window; past tests under these conditions have produced similar response patterns. The variable X_2 counts the advertisements that lead to purchases so she can decide whether to expand the campaign.
3. A Colombian streaming startup needs its first paying subscriber from a new visitor segment before expanding its promotion. It presents the same price to successive eligible visitors and ends the test when someone first subscribes; recent visitors in the segment have responded comparably to this offer. The variable X_3 records how many offers are presented through that first subscription, informing the cash needed to enter the segment.
4. An independent Bogotá animator submits one quote for an advertising project from an overseas client. The quote will either be accepted, filling the production calendar with foreign-currency income, or declined, leaving the month open for local commissions. The variable X_4 records which outcome occurs so the animator can schedule work and plan cash flow.

Questions/Tasks.

C1 Characterise each random variable and assign it to the appropriate probability distribution.

C1 - Characterise the random variables.

All four variables are discrete.

1. X_1 is **Poisson**: it counts separate enquiries within a fixed two-day interval at an approximately stable rate.
2. X_2 is **binomial**: it counts purchases in exactly 25 independent binary advertising trials with constant success probability.
3. X_3 is **geometric**: it counts independent offers through the first subscription when the success probability is constant.
4. X_4 is **Bernoulli**: a single quote has exactly two possible outcomes, accepted or declined.

2.3 Sustainable Trade Operations

Consider the following four situations.

1. A Bogotá sustainable-fashion brand inspects one returned jacket before settling the customer's refund. The inspection will classify it either as repairable for resale or as unsuitable for repair and destined for recycling. The variable X_1 records that classification because it changes both the refund reserve and the value recoverable from inventory.
2. A Colombian cacao exporter reviews a batch of 100 customs declarations before dispatch. Errors are discovered as separate omissions or mismatches, and in recent batches prepared by the same process they have appeared sporadically across the paperwork without a predictable position. The variable X_2 counts the errors in the batch so the exporter can contract enough specialist review and avoid costly delays.
3. A Colombian crafts cooperative sends catalogues to exactly 18 international retailers to plan an export launch. All receive the same wholesale terms, and for each retailer the cooperative records whether a sample is requested; comparable retailers contacted in recent rounds have shown similar responses. The variable X_3 counts the sample requests so the cooperative can buy materials and reserve artisan time without overproducing.
4. A Bogotá solar-equipment importer reviews prospective suppliers until it finds the first one meeting its certification rule. The same due-diligence process is applied to one candidate after another, and recent candidates drawn from the approved trade directory have shown comparable qualification records. The variable X_4 records how many suppliers are checked through the first qualifying one, helping forecast sourcing time and review costs.

Questions/Tasks.

C1 Characterise each random variable and assign it to the appropriate probability distribution.

C1 - Characterise the random variables.

All four variables are discrete.

1. X_1 is **Bernoulli**: one jacket produces exactly one of two outcomes, repairable or not repairable.
2. X_2 is **Poisson**: it counts separate errors in a fixed batch under an approximately stable rate.
3. X_3 is **binomial**: it counts sample requests across 18 fixed, independent binary trials with equal success probability.
4. X_4 is **geometric**: independent supplier checks continue until the first qualifying supplier, with constant success probability.

2.4 Finance, Tourism, and Mobility

Consider the following four situations.

1. A digital lender reviews Colombian microbusiness applications until it approves the first loan for a pilot product. Analysts apply the same lending rule to successive applicants from one eligibility list and stop after the first approval; earlier lists assembled in this way have shown comparable approval experience. The variable X_1 records how many applications are reviewed through that approval so the lender can budget underwriting time and acquisition cost.
2. A Bogotá bicycle-share operator monitors breakdown reports on a fixed 20-kilometre network segment for one week. Each report concerns a separate incident, and operational records for ordinary weeks show incidents scattered through the segment and week without a recurring timetable. The variable X_2 counts the reports in that period so the operator can position mechanics and spare parts without excess inventory.
3. A Colombian tourism platform presents the same local-experience package to exactly 60 eligible travellers. For every traveller it records whether the package is purchased, and previous groups recruited through the same channel have responded comparably. The variable X_3 counts the buyers so the platform can reserve enough Bogotá guides without making unnecessary advance payments.
4. A currency-management app monitors one scheduled dollar payment for a Colombian importer buying production inputs. By the deadline, the payment will either have executed or remain unexecuted; those are the only operational outcomes used by the app. The variable X_4 records which occurs because a missed deadline could interrupt the shipment and require costly short-term finance.

Questions/Tasks.

C1 Characterise each random variable and assign it to the appropriate probability distribution.

C1 - Characterise the random variables.

All four variables are discrete.

1. X_1 is **geometric**: equal-probability independent reviews continue until the first approval.
2. X_2 is **Poisson**: it counts separate reports over a fixed network segment and week at an approximately stable rate.
3. X_3 is **binomial**: it counts buyers in exactly 60 independent binary trials with constant purchase probability.
4. X_4 is **Bernoulli**: one scheduled payment has exactly two outcomes, on time or not on time.

3 Continuous Probabilities

Problems:

3.1 Events, Exports, and Platforms

Consider the following four situations.

1. A Bogotá concert organiser must fit an artist's sound check between venue setup and opening the doors. Production records place the duration X_1 from 30 to 70 minutes, with 45 minutes occurring most often; durations become less frequent as they approach either operational extreme. This pattern helps the organiser reserve enough venue time without routinely paying unnecessary overtime.
2. A flower exporter near Bogotá purchases sleeves and cartons before filling overseas orders. On its mature production line, stem length X_2 clusters around 60 centimetres, with roughly balanced departures above and below that typical length and progressively fewer stems far from it. The exporter uses this behavior to select packaging sizes that protect saleable output and limit waste.
3. A Bogotá ride platform is testing promotional credits while keeping each offer within COP 5,000 to COP 15,000. For the credit amount X_3, past randomised campaigns give no reason to expect one part of that range more often than another. Sampling offers in this way lets the platform compare demand across the budget range before choosing a standard incentive.
4. A marketplace connecting Colombian video creators with advertisers must budget one short-video placement priced from COP 100,000 to COP 500,000. Its transaction history shows that the observed price X_4 becomes steadily more common as the amount rises across that range, without a reversal before the upper limit. The marketplace uses this pattern to set campaign deposits and avoid a working-capital shortfall.

Questions/Tasks.

C1 Characterise each random variable and assign it to the appropriate probability distribution.

C1 - Characterise the random variables.

All four variables are continuous.

1. X_1 is **triangular**: it has bounded minimum and maximum values, an interior most likely value, and density that rises then falls linearly.
2. X_2 is **normal**: lengths are approximately symmetric around a centre and become progressively rarer with distance from it.
3. X_3 is **uniform**: it is bounded and every value in the interval is equally plausible.
4. X_4 is **linear**: its density is one increasing straight line across the full bounded support.

3.2 Food, Finance, and Content

Consider the following four situations.

1. An artisan bakery supplying Bogotá restaurants monitors loaf weight X_1 because portions affect ingredient margins and clients' menu costs. Mature production records concentrate near 800 grams, with similar variation on either side and increasingly few loaves at large departures from that amount. The bakery uses this pattern to plan flour purchases and reduce waste without under-serving clients.
2. An independent worker in Engativá uses a finance app to move part of each day's earnings into savings between 8:00 and 10:00 p.m. For the transfer time X_2, processing history offers no reason to anticipate one part of the two-hour window more often than another. The worker relies on that window to keep funds available for evening expenses while the app spreads its workload.
3. A Bogotá video creator is preparing a fixed-price proposal whose margin depends on editing effort and collaborator fees. The total editing cost X_3 cannot practically fall below COP 900,000 or exceed COP 1,800,000, and similar projects most often cost COP 1,200,000; costs appear less often as they approach either limit. The creator uses these estimates to quote a fee that protects the project margin.
4. A Colombian digital bank studies identity-check duration X_4 to size servers without overspending on capacity. Verifications take from zero to five minutes in its operating records, and durations become steadily less common from the shortest end to the longest, with no change in that downward pattern. The bank uses the behavior to plan infrastructure while controlling account-opening delays.

Questions/Tasks.

C1 Characterise each random variable and assign it to the appropriate probability distribution.

C1 - Characterise the random variables.

All four variables are continuous.

1. X_1 is **normal**: weights are approximately symmetric about a centre and taper in frequency on both sides.
2. X_2 is **uniform**: every instant in a bounded two-hour interval is equally plausible.
3. X_3 is **triangular**: it has stated bounds and an interior mode, with linear rise and fall in density.
4. X_4 is **linear**: the density decreases in one straight line throughout its bounded support.

3.3 Tourism, Logistics, and Design

Consider the following four situations.

1. A Colombian craft brand must finance materials and artisan wages before an overseas buyer pays for the next order. Recent small export orders range from COP 1 million to COP 11 million, and the observed order value X_1 becomes steadily less common as value rises across that range, without reversing direction. The brand uses this behavior to arrange working capital while avoiding an unnecessarily large credit line.
2. A La Candelaria tour operator coordinates guides with timed reservations at Bogotá food businesses. Route duration X_2 has a practical minimum of two hours and maximum of four hours, and three hours occurs most often; observations thin out toward both limits. The operator uses this experience to protect reservations from

delays without routinely paying guides for excessive standby time.

3. A logistics broker handling an import through Colombia must dispatch a truck after customs release but before a reserved warehouse slot. Release time X_3 will occur between 9:00 a.m. and noon, and comparable clearances give the broker no reason to favour any part of that window. This information guides a dispatch choice that limits both truck waiting charges and late-arrival penalties.
4. A Bogotá footwear designer is choosing a viable width range before funding a Colombian production run. In a large target-market sample, foot width X_4 is concentrated near a typical measurement, with comparable variation on either side and progressively fewer observations at more distant measurements. The designer uses that pattern to cover most buyers without carrying too many costly variants.

Questions/Tasks.

C1 Characterise each random variable and assign it to the appropriate probability distribution.

C1 - Characterise the random variables.

All four variables are continuous.

1. X_1 is **linear**: its bounded-support density decreases along one straight line from one endpoint to the other.
2. X_2 is **triangular**: the bounded duration has a most likely interior value and two linear density segments.
3. X_3 is **uniform**: every time in the stated bounded window is equally plausible.
4. X_4 is **normal**: measurements are approximately symmetric around a centre and less frequent farther from it.

3.4 Technology, Commerce, and Sustainability

Consider the following four situations.

1. A Bogotá cloud startup must reserve infrastructure for a client whose storage use can range from zero to 20 terabytes. Contract records show that the observed demand X_1 becomes steadily more common as use rises across that range, without turning downward before the contractual cap. The startup uses this behavior to buy enough capacity while limiting the cost of unused resources.
2. A circular-economy venture in Kennedy bids for used appliances while preserving margin for parts, labour, and warranties. The resale price X_2 has a practical minimum of COP 400,000 and maximum of COP 900,000, with COP 650,000 most common in the venture's records; prices occur less often nearer either extreme. This experience helps the buyer set a prudent inventory bid.
3. An international e-commerce platform adjusts Colombian sellers' peso receipts for exchange-rate movements. For the adjustment X_3, treasury policy limits the change to between negative 1.5 percent and positive 1.5 percent, and historical transactions provide no reason to expect one part of that range more often than another. Sellers use this information to test margins and choose an adequate currency-risk buffer.
4. A Colombian coffee exporter monitors moisture percentage X_4 because unsuitable storage can reduce inventory value before shipment. Readings from bags handled under the usual process gather around 11 percent, show similar behavior below and above that typical level, and become increasingly scarce farther away. The exporter uses the pattern to prioritise inspections and protect export quality.

Questions/Tasks.

C1 Characterise each random variable and assign it to the appropriate probability distribution.

C1 - Characterise the random variables.

All four variables are continuous.

1. X_1 is **linear**: its density increases along one straight line across a bounded support.
2. X_2 is **triangular**: stated bounds, an interior most likely price, and linear rise and fall identify it.
3. X_3 is **uniform**: all real values in a bounded interval are equally plausible.
4. X_4 is **normal**: moisture values are approximately symmetric around a central value and rarer farther away.